

Probability Distributions

How the values of a random variable are distributed.

- Uniform
- Binomial
- Normal

1 Poisson Distribution

Model the no. of times an event occurs in a fixed interval of time or space.

- Events happen independently
- Constant average rate.

Σ $\hookrightarrow$ No. of customers entering a store per hour

Σ No. of emails received in a minute

Σ No. of accidents happening at a traffic junction

Σ No. of defects in a manufacturing unit

Key conditions

- $\underline{=}$ Events occur independently
- $\underline{=}$ Events occur randomly
- $\underline{=}$ The average rate is constant

Key Property $\lambda = \text{average no. of events}$

$$P(X=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

X = Random variable

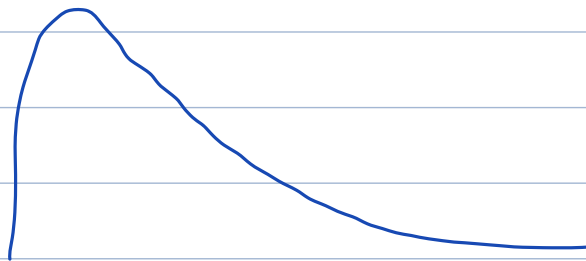
k = no. of occurrences

λ = average rate

$e = 2.718$

Mean = Variance

Shape of distribution. $\lambda = \text{small}$



When do we use Poisson distribution

n is large
 p is very small

$$np = \lambda \text{ (finite constant)}$$

$p \rightarrow$ success $\rightarrow$ rare

$\rightarrow$ No. of failed operations by a doctor

$\rightarrow$ No. of misprints in a book

$\rightarrow$ Errors in data transmission

Exponential Distribution

Continuous prob distribution that models the time between consecutive events, the events occur randomly at a constant average rate

λ = No. of customers per hour

$\uparrow$
The time between two customers entering the store

Event	Time	Gap
Cust 1 arrives		
Cust 2 arrives		3 min later
Cust 3 arrives		1 min later
Cust 4 arrives		7 min later

Key Parameter λ = average rate of events

If $\lambda = 2$ events per hour

PDF $f(x) = \lambda e^{-\lambda x}$

$x \rightarrow$ wait time

$\lambda \rightarrow$ rate parameter

$e = \underline{\underline{2.718}}$

A call center receives 2 calls per minute.
What is the prob that the next call comes after 1 minute?

$x = 1, \lambda = 2$

$f(x) = \lambda e^{-\lambda x}$

$f(1) = 2 \cdot e^{-2 \cdot 1}$

$= 2 \cdot e^{-2}$

$f(1) = 2 \cdot (2.718)^{-2} = 2 \cdot (0.135) = 0.27$

Properties

$$\text{Mean} : \frac{1}{\lambda}$$

$\lambda = 2$ events per hour

On avg, a new event happens $\frac{1}{2}$ hours

$$\text{Variance} : \frac{1}{\lambda^2} \Rightarrow \frac{1}{2^2} = 0.25$$

$$\text{SD} = \text{Mean} = \frac{1}{\lambda}$$

$\lambda \rightarrow \text{high}$

Events happen more freq
Waiting time is small

$\lambda \rightarrow \text{low}$

Events are rare
Waiting time is long

Geometric Distribution

No. of trials needed to get the first success

Ex Tossing a coin until first head

Key Idea

Succ. prob = p

Failure prob = $1 - p = q$

$$P(X=x) = (1-p)^{x-1} \cdot p$$

Ex Coin Toss

Prob of head = $p = 0.5$

Prob first head comes on 3rd toss.

$$\begin{aligned} P(X=3) &= (1-0.5)^{3-1} \cdot 0.5 \\ &= (0.5)^2 \cdot 0.5 \\ &= 0.125 \end{aligned}$$

Ex

Prob of solving $p = 0.2$
First success on 4th trial

$$\begin{aligned} P(X=4) &= (1-0.2)^{4-1} \cdot 0.2 \\ &= (0.8)^3 \cdot 0.2 \end{aligned}$$

t-Distribution

"What is the avg height of all students in a college?"

Population $\rightarrow$ Sample (10-15)

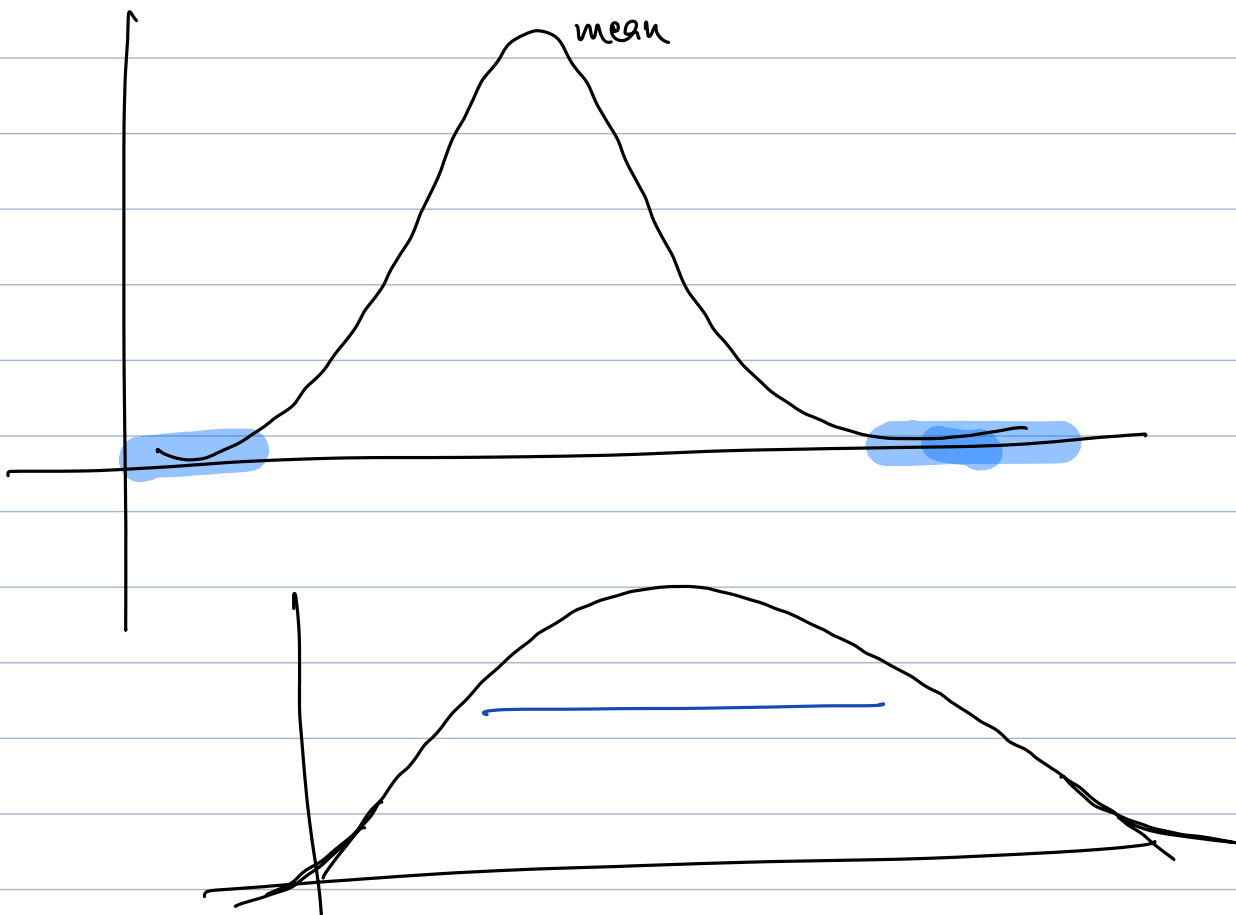
$\Rightarrow$ Sample mean ($\bar{x}$)

$\Rightarrow$ Sample standard Deviation (s)

Small sample $\rightarrow$ less data

Using s instead of $\sigma \rightarrow$ extra uncertain

Our estimates may be wrong



Degree of freedom

Z-score

Mean
50

Value
80

+ Z
- Z

Normal

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

t-distr

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Sample mean

Pop mean

Sample SD

Sample size

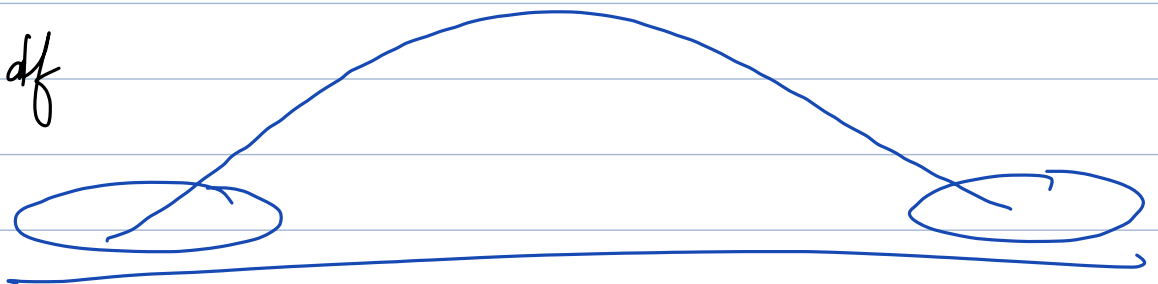
Degrees of Freedom

$$df = n - 1$$

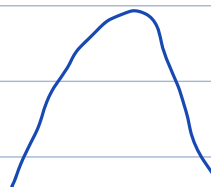
⇒ sample mean of n values

(n-1) → n

Smaller df



Larger df



Large t -value : against the hypothesis

Smaller t -value : sample is close to the assumed mean

⇒ increase the sample size → normal

Chi-square Distribution (χ^2)

continuous prob distribution -

- Categorical data analysis
- Comparing observed vs expected values

— works on counts / frequencies

where is it used?

1. Test of Independence

Check if 2 categorical variables are related

→ Gender Vs Product Preference

→ Course Vs Placement Outcome

2. Goodness of Fit Test

Checks how well data fits an

expected distribution

- Is a die fair?
- Does data follow a distribution
- Are customer choices evenly distributed?

Compare Observed values Vs Expected Value

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$O \approx E \rightarrow$ small value $\rightarrow$ data fits expectation
 O and E are very different $\rightarrow$ mismatch



Shape $\rightarrow df = n - 1$

small
large

Highly
more

skewed
symmetric

Distribution	Used for	Example
Uniform	Equal prob events	Dice roll
Binomial	Fixed trials / Success	No. of heads
Normal	Continue data around mean	Marks
Poisson	Count of events in interval	Customers per hour
Exponential	Time b/w events	Time between arrival of customers
Geometric	Trials until the first success	Coin tosses until head
t-dis.	Small sample	Hypothesis Testing
Chi-square	Categorical data testing	Independence tests

Aerofit Case-Study Deadline
is 27th March